

Midterm 2 Part a

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I. QUESTIONS (40 POINTS)

1. How many input and output units (total) are there for a linear perceptron with $d = 7$ for binary classification?
 - a) 6
 - b) 7
 - c) 8
 - d) 9
 - e) 10
2. How many weights are there for a linear perceptron with $d = 6$ for binary classification?
 - a) 5
 - b) 6
 - c) 7
 - d) 8
 - e) 9
3. How many input and output units (total) are there for a linear perceptron with $d = 5$ for $K = 3$ classification?
 - a) 8
 - b) 9
 - c) 10
 - d) 15
 - e) 24
4. How many weights are there for a linear perceptron with $d = 3$ for $K = 4$ classification?
 - a) 8
 - b) 15
 - c) 16
 - d) 20
 - e) 24
5. How many input, hidden, output units (total) are there for a multi layer perceptron with $d = 8$, hidden node count 5, for binary classification?
 - a) 9
 - b) 12
 - c) 16
 - d) 22
 - e) 32
6. How many weights are there for a multi layer perceptron with $d = 4$, hidden node count 2, for binary classification?
 - a) 9
 - b) 13
 - c) 16
 - d) 22
 - e) 32
7. How many input and output units (total) are there for a multilayer perceptron with $d = 3$, hidden node count 2, for $K = 3$ classification?
 - a) 10
 - b) 14
 - c) 20
 - d) 32
 - e) 34

8. How many weights are there for a multilayer perceptron with $d = 3$, hidden node count 3, for $K = 5$ classification?

- a) 10
- b) 14
- c) 20
- d) 32
- e) 34

9. Which of the following functions can not be learned by linear perceptron?

- a) $3x + 4$
- b) 5
- c) $x^2 + 1$
- d) $4x - 2$

10. How many separate hyperplanes (discriminant functions) are defined when $K = 20$ in linear classification?

- a) 2
- b) 5
- c) 9
- d) 10
- e) 20

11. How many separate hyperplanes (discriminant functions) are defined when $K = 5$ in pairwise linear classification?

- a) 2
- b) 5
- c) 9
- d) 10
- e) 45

12. What is the value of sigmoid function when $x = 0$?

- a) 1
- b) -1
- c) 0
- d) 0.5

13. What is the derivative of the sigmoid function y ?

- a) y
- b) $1 - y$
- c) $y(1 - y)$
- d) $\ln y$
- e) $\frac{1}{y}$

14. What is the derivative of the tanh function y ?

- a) y
- b) $1 - y^2$
- c) $y(1 - y)$
- d) $\ln y$
- e) $\frac{1}{y}$

15. If there are 4 classes, which of the following vector can be the output of the softmax function?

- a) [0.3 0.3 0.3 0.1]
- b) [0.3 0.3 0.4 0.2]
- c) [0.4 0.3 0.1 0.1]
- d) [1.0 1.0 1.0 1.0]
- e) [0.5 0.5 0.5 0.5]

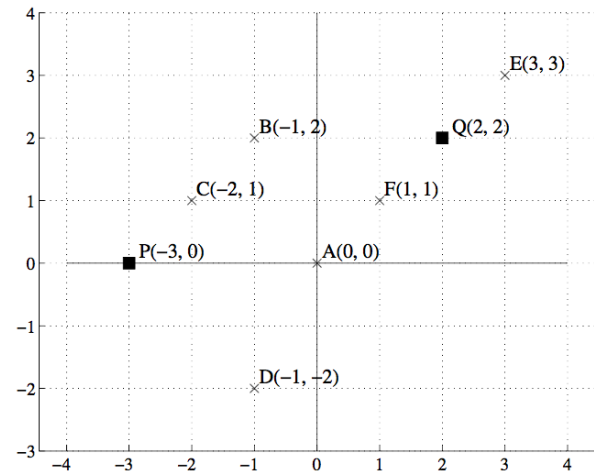
16. Which of the following is not a way of generating diverse learners?

- a) Use different learning algorithms
- b) Use the same algorithm but with different hyperparameters
- c) Use different representations of the same input
- d) Use different subsets of the training set to train algorithms
- e) All are correct

17. Which of the following is an example for multistage model combination technique?

- a) Adaboost
- b) Stacking
- c) Cascading
- d) Voting

Suppose we are given the following dataset. In the following questions 18-21, we will do k -means clustering to cluster the points A, B ... F into 2 clusters. The current cluster centers are P and Q. Let say the distance function used in cluster determination is absolute distance instead of euclidian distance.



18. Which points are assigned to cluster center P?

- a) A, B, C, D
- b) B, C, D
- c) B, C
- d) A, C, D
- e) C

19. Which points are assigned to cluster center Q?

- a) E, F
- b) E
- c) A, F, E
- d) B, F, E
- e) F

20. What does cluster center P get updated to?

- a) -1, -1/3
- b) -4/3, 1 / 3
- c) -3 / 2, 3 / 2
- d) -3 / 5, 2 / 5
- e) -2, 1

21. What does cluster center Q get updated to?

- a) 2, 2
- b) 3, 3
- c) 4 / 3, 4 / 3
- d) 3 / 4, 6 / 4
- e) 1, 2

Consider learning a decision tree that you could use to judge whether or not you will like a given restaurant. Assume you have chosen to use the following two features to describe restaurants, with the possible values shown.

Price $\in \{\text{Low, Med, High}\}$ Type $\in \{\text{Hamburgers, Pizza, Fish, Vegetarian}\}$

Assume decision algorithm is given the following set of classified training examples.

P	T	Like
L	H	+
L	V	+
M	F	-
M	V	+
H	P	-

$\log 2 = 1, \log 3 = 1.58, \log 4 = 2, \log 5 = 2.32.$

22. Which of the following feature will be selected feature at the root node?

- Price
- Type

Assume you are given the following three features of TV shows, with the possible values shown, and wish to learn how to predict future top-10 shows.

$\text{Type} \in \{\text{Comedy, Drama, News, Sports}\}$

$\text{Location} \in \{\text{LA, NYC, Various}\}$

$\text{Duration} \in \mathbb{Z}$

Assume decision tree construction algorithm is used on the following examples.

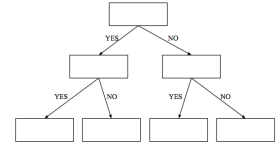
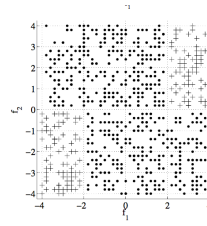
Type	Location	Duration	ClassInfo
C	NYC	30	+
C	LA	40	-
N	Var	60	+
N	LA	60	+
S	Var	40	-
D	LA	40	-

23. Which of the following feature will be selected feature at the root node?

- Type
- Location
- Duration

24. Which of the following is an example for local multiexpert model combination technique?

- Voting
- Adaboost
- Cascading
- Mixture of experts



25. What will be the first three nodes of the decision tree that will be constructed for the data above?

- $f_1 > 2, f_2 > 0, f_1 > -2$
- $f_1 > 2, f_1 > -2, f_2 > 0$
- $f_1 > -2, f_2 > 0, f_1 > 2$
- $f_1 > -2, f_1 > 2, f_2 > 0$
- $f_2 > 0, f_1 > 2, f_1 > -2$

Given 5 learners and their posterior probabilities on 4 classes in the table below:

	C_1	C_2	C_3	C_4
l_1	0.1	0.2	0.3	0.4
l_2	0.0	0.5	0.5	0.0
l_3	0.25	0.25	0.25	0.25
l_4	0.2	0.3	0.2	0.3
l_5	0	0.8	0.1	0.1

26. What is the output of the sum rule for each class?

- [0.11 0.41 0.27 0.21]
- [0.1 0.3 0.25 0.25]
- [0.0 0.2 0.1 0.0]
- [0.25 0.8 0.5 0.4]
- [0.0 0.06 0.0075 0.0]

27. What is the output of the median rule for each class?

- [0.11 0.41 0.27 0.21]
- [0.1 0.3 0.25 0.25]
- [0.0 0.2 0.1 0.0]
- [0.25 0.8 0.5 0.4]
- [0.0 0.06 0.0075 0.0]